

**COURTSHIP RITUAL**

A male frigatebird inflates his throat pouch to attract the attention of a female. Birds have good color vision, which explains why males often have striking plumage.

Fertilization

Marking the start of a new life, fertilization occurs when a male sperm and female egg cell fuse. In animals, it takes place in one of 2 ways: either outside or inside the female's body.

External fertilization is used by many animals that either live permanently in water or return to it to breed. In the simplest version of this process—seen in static invertebrates such as corals—vast numbers of the male and female sex cells are shed into the water, where they mingle so that fertilization can occur. A more advanced version of this, shown by animals that can move around, such as frogs, involves 2 partners pairing up. Although they appear to mate, fertilization nevertheless takes place in the water rather than inside the female's body.

External fertilization does not work on land because sex cells soon dry out and die when exposed to air. Most terrestrial animals therefore use internal fertilization. In general, this involves the male injecting sperm into the female. However, some terrestrial animals, such as salamanders and newts, do not copulate. Instead, the male deposits a package of sperm (a spermatophore) near the female; she then collects it with her reproductive organ so that internal fertilization can take place.

**INTERNAL FERTILIZATION**

Like all insects, flat-footed bugs have to pair up so that the female's eggs can be fertilized. Mating takes several hours.

EXTERNAL FERTILIZATION

Corals release their sex cells into the water. They are triggered to do this by the changing phases of the moon—a natural clock that many animals use to synchronize their breeding behavior.

Starting life

Most animals—apart from the ones that use asexual reproduction—start life as a single fertilized egg cell. If the egg has been fertilized externally, it will already be outside the mother's body, perhaps drifting in the sea or glued to seabed plants or sand. If the egg has been fertilized internally, it will either be laid, to hatch afterward, or it will be retained inside the mother while it begins to develop into a young animal.

The degree of development that takes place at any one stage varies from one type of animal to another. Oviparous species, such as birds, lay their fertilized eggs before fetal development begins. In birds, development is often deferred for several

LIVE BIRTH

Animals that give birth have relatively few offspring, but they often show a high level of parental care. This female seal will look after her cub for many weeks.



more days until the clutch is complete; it begins as soon as the parent starts to incubate the eggs. Ovoviviparous species, which include many reptiles and sharks, incubate their eggs internally, “giving birth” at the moment when the eggs are about to hatch. Viviparous species—which include virtually all mammals, as well as some reptiles, amphibians, and fish—give birth to live young.

Metamorphosis

All animals change shape as they grow and develop. In some, the changes are gradual and relatively minor, but in others, they are so far-reaching that the animal is completely transformed. This transformation is called metamorphosis. It allows animals to live in different ways—and often in different habitats—during their young and adult lives.

LIVING TOGETHER

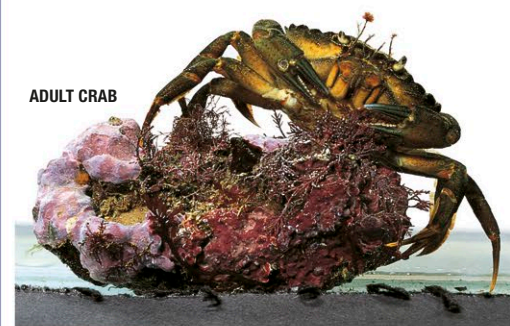
Many animals associate with their own kind but continue to lead independent lives. However, eusocial species, such as termites, ants, and bees, form permanent groups or colonies in which just one member—the queen—produces all the colony's young. As a result, the colony's members are closely related and in many ways behave like a single organism. The great success of the system is apparent in the fact that these insects are among the most numerous on earth.

**QUEEN TERMITE**

Hidden deep inside a termite nest, where she is attended by her workers, a queen termite lays up to 30,000 eggs a day. In her complete dependence on the workers for food, she represents the ultimate form of reproductive specialization.

Although metamorphosis is most common in invertebrates, it does occur in amphibians and some fish. Animals that undergo the metamorphic process spend the early part of their lives as larvae. In the sea, larvae often drift near the surface as part of the plankton, and because they are carried far and wide they play an important role in helping their species spread.

In the insect world, metamorphosis occurs in 2 ways. Incomplete metamorphosis, shown by grasshoppers and bugs, involves a series of gradual changes that are made as the young insect, or nymph, matures. Complete metamorphosis—shown by butterflies, beetles, and flies—involves more drastic changes, which occur during a resting stage, called pupation, when the body is broken down and rearranged.

**CRAB EGGS****ZOEA LARVA****ADULT CRAB****FROM LARVA TO ADULT**

Like most crustaceans, shore crabs start life as eggs (top). The eggs hatch to produce the first larval stage, called a zoea (center), which floats in plankton. This changes into a megalopa larva, then sinks to the seabed and finally changes into an adult crab.